
RETHINKING THE UI OF GENUI: A TALE OF TWO DESIGNS

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Xiang ‘Anthony’ Chen*

HCI Research, UCLA
Los Angeles, California, United States

Savvas Dimitrios Petridis

Google DeepMind
New York, New York, United States

Tian Deng

Cloud AI, Google
Mountain View, California, United States

Humad Bari

HCI Research, UCLA
Los Angeles, California, United States

Ruofei Du

Google XR
San Francisco, California, United States

Yang Li

Google DeepMind
Mountain View, California, United States

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ABSTRACT

GenUI is an emergent class of AI tools that use large models (LM) to generate UI mock-ups based on users’ high-level descriptions, promising to democratize UX design exploration for broader audience. Most GenUI designs to-date tend to inherit the conventions of conversational LMs (*e.g.*, ChatGPT and Gemini), where a user describes their design needs primarily via an unstructured prompt, and the tool then takes a depth-first approach, delving into the design right away and producing a high fidelity prototype. In this research, we rethink how well this unstructured, depth-first, and high-fidelity GenUI design can support the important early-stage, 0-to-1 design exploration. To probe this question, we propose a contrastive design with structured input, breadth-first exploration, and low-fidelity generation. We then conducted a comparison study with 24 UX designers and product managers who conducted mini design exploration exercises using an existing and our contrastive GenUI tools. Findings reveal participants’ perceived benefits and trade-offs of the two GenUI designs: (i) Structured input surfaces key facets but requires more work, raising entry barriers to start exploration; (ii) Breadth-first workflow reveals more possibilities, but previewing UX ideas spanning many screens remains hard; and (iii) Though low fidelity has value, professionals favor high fidelity—it fits practice, and GenAI heightens fidelity expectations. We conclude with design implications for GenUI and similar AI-powered creativity support tools.

Keywords GenUI · UX Design · Generative AI · Design Tools

1 Introduction

Pre-trained large models can now enable **Generative User Interfaces (GenUI)**—generating UI screen mockups from high-level inputs (*e.g.*, a textual description), promising to democratize UX design exploration for broader audience. One major promise of GenUI is supporting the important early-stage, “0-to-1”² ideation Chen et al. (2025), which often

*Corresponding author: xac@ucla.edu

²There are two distinct types of design needs that GenUI tools can support: “0-to-1” vs. “n-to-(n+1)”, where the former is about exploring a new feature or product from scratch and the latter is about adding or changing features on an existing product.

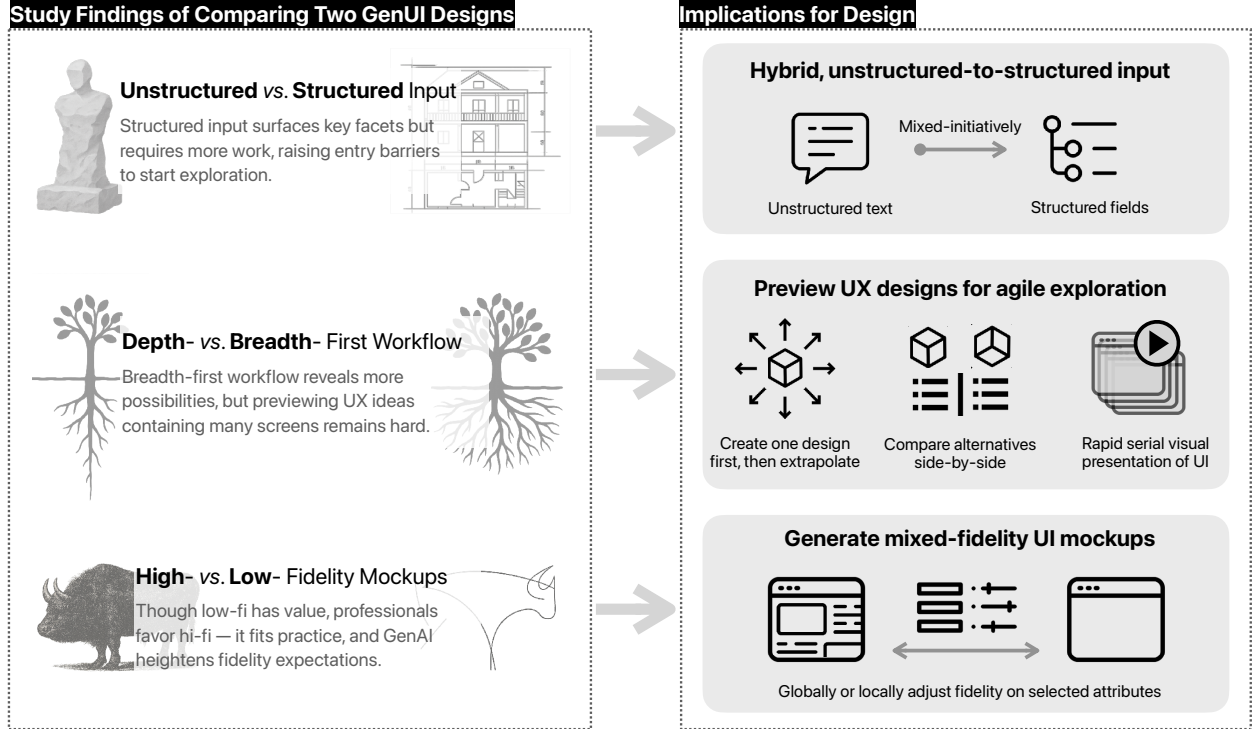


Figure 1: **We conducted a study to understand two contrastive designs of GenUI**—tools that employ large models to generate UI mockups: (i) Input: a conventional open-ended, unstructured prompt vs. a structured form that guides users to provide key information; (ii) Workflow: delving into a design right away (depth-first) vs. starting with exploring a design space (breadth-first); (iii) Output: generating polished, high-fidelity vs. rough, low-fidelity UI mockups.

involves both UX designers (UXD) and product managers (PM) Chamberlain et al. (2006); Melegati et al. (2024). For example, in a design sprint Banfield et al. (2015), both UXD and PM can use GenUI to translate requirements and user journey into specific design ideas. GenUI targets these early-stage 0-to-1 stakeholders—primarily PM and UXD—whose responsibilities diverge later in the process but share many common activities in the early phase.

Currently, the UI of most GenUI tools —e.g., Uizard Uizard (2025), Visily Visily (2025), Luny AI Luny AI (2025), Figma Make Figma, Inc. (2025), and Stitch Google (2025b)— typically follows the paradigm of chat-based AI (e.g., ChatGPT and Gemini) where a user interacts with a GenUI agent primarily via a thread of unstructured text prompts. These GenUI tools often pursue a “depth-first” approach, attempting to generate a complete high-fidelity UI mock-up (e.g., in HTML/CSS) in one go, followed by more prompts for refinement.

While this UI paradigm of GenUI remains a popular approach, it comes with potential trade-offs. First, an unstructured text prompt allows for easy and free expression, yet information provided by users (especially non-designers) might be incomplete and ambiguous Brade et al. (2023); Morris (2024); Suh et al. (2024); Riche et al. (2025), leading to the model’s suboptimal interpretation of their design needs. Second, going straight from prompts to UI generation presents a fast result but might overlook the discover phase Design Council (nd)—divergent Goldschmidt (2016) exploration of the underlying design space Suh et al. (2024) as an important purpose of prototyping Lichter et al. (1994); Houde and Hill (1997); Dow et al. (2011); Tohidi et al. (2006); Lim et al. (2008). Third, generating a high-fidelity UI mock-up matches what conventional design tools (e.g., Figma) can offer, yet its attention to fine details risks prematurely locking the design into a specific direction Virzi (1989); Landay and Myers (1995); Buxton (2010); Greenberg and Buxton (2008). While GenUI systems are emerging in HCI Lu et al. (2025); Leung et al. (2025); Yuan et al. (2025); Kolthoff et al. (2025), a systematic investigation of these three key design choices remains absent from prior literature.

In this research, we rethink the UI of GenUI tools by juxtaposing the existing paradigm against a contrastive approach along the following dimensions, encompassing the end-to-end process of interacting with a GenUI tool:

- **Input:** Instead of unstructured text prompts, we build on prior work Owen and Roberts (2024); Lu et al. (2024a); Shokrizadeh et al. (2025) and instrument a *structured* input format that employs the critical user journey (CUJ) method Vachon (2025), guiding users to specify the context, user, goal, and tasks related to their design needs.

- **Workflow:** Instead of going depth-first, we employ a *breadth-first* workflow, found in numerous creativity support tools Tohidi et al. (2006); Lim et al. (2008); Suh et al. (2024); Chang et al. (2018); Ma et al. (2025), that starts with generating a design space for users to see and explore multiple design alternatives before delving into one to generate the UI mockups.
- **Output:** Instead of high fidelity, we opt to generate *low-fidelity* UI mockups that are less polished, yet a lack of fine details could in turn reduce information load while allowing more focus on the high-level design ideas.

Our central research question is: *How do these design choices of input, workflow, and output affect the usage patterns, benefits, and trade-offs of GenUI tools for early-stage design exploration?*

To investigate this question along the three dimensions, we created MINDSPAN—a proof-of-concept GenUI tool that implemented the core design ideas of structured input, breadth-first workflow, and low-fidelity output. Using MINDSPAN as a probe, we then conducted hour-long workshops with a total of 24 professionals from the same software company, including both PM and UXD as they are the two key players in the early ideation phase supported by GenUI. Following a within-subject design, each participant used two GenUI tools (ours and Stitch Google (2025b)—a state-of-the-art GenUI tool representing the existing paradigm) to conduct two 0-to-1 design tasks. After each task, participants filled out survey questions to rate the two GenUI designs’ input, workflow, and output. Within two workdays after the workshop, we conducted one-on-one semi-structured interviews with each participant to further discuss how they compared the two GenUI tool designs.

Our key findings are as follows:

- While structured input helped participants think through the many key facets of UX design (user, goal, and tasks), it also increased the barrier to entry compared to unstructured input, calling for a hybrid approach and more support for a “warm start”, especially providing examples to build off of.
- The breadth-first workflow outperformed the depth-first approach in challenging participants’ assumptions and led them to discover otherwise overlooked design possibilities. However, one main challenge is previewing a UX design idea when exploring many, as each idea itself also consisted of multiple screens of information to support multiple task flows.
- Despite the many reasons to favor low fidelity for design exploration, participants still overwhelmingly preferred high fidelity because it better fits their professional UX practices (including a recent surge of “vibe coding” product-like mockups). Such a “high-fidelity mindset”, combined with expectations built up via the breadth-first exploration, amplified participants’ disappointment when presented with low-fidelity mockups.

Building off of these findings, we discuss implications for designing GenUI and other AI-powered creativity support tools (illustrated in Figure 1), including (i) A hybrid, mixed-initiative, unstructured-to-structure input to ease into GenUI exploration. (ii) Previews of UX designs to support agile exploration, e.g., create one design first, then extrapolate; compare alternatives side-by-side; and rapid serial visual presentations of UI screens. (iii) Generating high-fidelity mockups by default but allowing for mixed-fidelity adjustments globally or locally on selected attributes.

The main contribution of this work lies in the findings and design implications from comparatively studying two paradigms of GenUI designs. Our results illuminate trade-offs across three key aspects—input, workflow, and output—that underpin the design exploration process in UX and beyond.

2 Related Work

GenUI employs AI to create prototypes (UI mockups) that support UI/UX work, which leads to our review of two related bodies of literature: the use of prototypes in UI/UX design process (§2.1) and studies of how designers use AI in their work (§2.2). Then, we focus on one specific type of AI for UX work—GenUI tools, reviewing the technological background and recent development.

2.1 Prototypes in UI/UX Design Process

The use of prototypes is a long-standing practice in software industry that predated modern UI/UX design process. Lichter *et al.* studied how industrial software teams adopted prototyping practices Lichter et al. (1994) and their findings surface three key attributes of prototypes, that (i) prototyping is part of the *evolutionary* approach of software development; (ii) prototyping aims at creating *early working versions* of a system or application; and (iii) prototyping provides a medium for *communication* among team members as well as users. Later work on UI/UX design process echoed and built on these three characteristics. Houde and Hill summarized three specific prototyping purposes:

understanding the *role* the system plays in its context of usage, exploring how the system looks and how users interact with it (*i.e.*, *look and feel*), and probing technical solutions to *implement* the system Houde and Hill (1997). Lim *et al.* proposed an anatomy of prototypes comprised of two dimensions: *filtering* certain aspects of the system design and *manifesting* the prototype in certain medium, which enables designers to take a systemic view of prototypes and prototyping practices Lim *et al.* (2008). Among all the above studies, one shared sentiment is that prototypes—whether created by humans or AI— should facilitate the iterative design and development of products, from exploring early ideas, to experimenting with implementation solutions, and to evolving an early working version to a product-ready artifact Lichter *et al.* (1994); Houde and Hill (1997); Lim *et al.* (2008). Our study is in part motivated by these findings, as we hypothesized that a breadth-first workflow and a low-fidelity output can facilitate the use of generated UI mockups to drive the iterative UI/UX process.

2.2 Studies of Designers’ Use of GenAI

AI is increasingly involved in UX design, with research exploring its integration into designers’ workflow Lu *et al.* (2024b); Wadinambiarachchi *et al.* (2025) (*e.g.*, the double-diamond model Design Council (nd)), its capacity to contextualize user data Salminen *et al.* (2019), and the recurring patterns and challenges defining its adoption Kneareem *et al.* (2023); Chaudhry (2024); Stige *et al.* (2023). New GenAI tools have emerged for UX design, such as generating user personas and design examples under various constraints Shokrizadeh *et al.* (2025) or support user journey mapping through inspirations and progressive discussions Mei *et al.* (2025). The GenUI tool we studied (MINDSPAN) similarly leverages GenAI to generate user personas and journeys, but extends this by integrating user-centered information to create UI mockups.

Recent empirical work reveals that while GenAI offers new opportunities for UX practice—such as generating personas, journey illustrations Uusitalo *et al.* (2024), and fostering divergent thinking through design exploration Khan *et al.* (2025)—it also introduces persistent challenges. These unresolved tensions span the end-to-end design process and include a lack of shared team practices and training Takaffoli *et al.* (2024), difficulties in hand-offs and cross-role communication Li *et al.* (2024), systemic collaborative hurdles Subramonyam *et al.* (2024b), and underlying skepticism from designers regarding GenAI’s promised potential Naqvi *et al.* (2025). Taken together, this body of work primarily examines GenAI tools as they currently exist; in contrast, our research challenges prevailing GenUI designs by proposing a contrastive approach that systematically explores the benefits and trade-offs of alternative designs along three GenUI design dimensions.

Other research focused on studying GenAI’s integration in UX design tools. Feng *et al.* developed a Figma plug-in to examine how LLMs translate requirements into concrete design artifacts Feng *et al.* (2025), and PromptInfuser links UI elements in Figma to model inputs and outputs to let designers simulate and probe AI functionality at design time Petridis *et al.* (2023). Closest to our work, Chen *et al.* (2025) studied multiple design and non-design roles using one commercial GenUI tool, reporting how GenUI supports each role, along with challenges and improvement opportunities. While Chen *et al.* (2025) addressed a broad range of GenUI issues, our studies delve deeper into three key aspects (input, workflow, and output) and are grounded in a contrastive study of two different GenUI designs rather than a single tool.

2.3 Overview of GenUI Tools

There are three families of GenUI tools based on different UI representations: (*i*) generating pixels of a UI screen image (*e.g.*, using Generative Adversarial Networks Goodfellow *et al.* (2014) or Diffusion models Ho *et al.* (2022) to provide design examples Mozaffari *et al.* (2022)); (*ii*) generating hierarchical data structure (*e.g.*, transforming an initial set of UI elements into a refined layout Li *et al.* ([n.d.]), and; (*iii*) generating source code (*e.g.*, using programming-by-demonstration to implement new interactive features Ritam Jyoti Sarmah *et al.* (2020) or transformers to generate tokens representing and rendering a UI Huang *et al.* (2021); Wu *et al.* (2024)).

Employing the code-generation approach, Latest developments of LMs have given rise to quite a few commercial tools, such as Uizard Uizard (2025), Visily Visily (2025), UX Pilot UX Pilot (2025), Luny AI Luny AI (2025), Figma Make Figma, Inc. (2025), Emergent AI (2025), Banani Banani (2025), Framer B.V. (2026), and Google’s Stitch Google (2025b). Most of these GenUI tools, such as Uizard, Stitch, Visily, and Luny AI, employ unstructured inputs (natural language, sketches, or screenshots), follow a depth-first generation strategy, and aim to produce high-fidelity, polished mock-ups in a single pass. For example, Uizard Uizard (2025) utilizes a transformer-based heuristic to map natural language tokens to a predefined design system schema, automatically generating layout structures, color variables, and font pairings that align with the semantic intent of the prompt. The engine executes a high-fidelity render by bypassing low-fidelity wireframing, directly instantiating complex UI components and multi-screen navigation flows from a single string of unstructured text. Notably, a few exceptions do depart from this design paradigm: UX Pilot introduces more

structured input via describing individual screens, while Div-idy Wikipedia (2025) emphasizes breadth-first exploration by decomposing prompts into modular components and reassembling them via a multi-stage pipeline.

3 GenUI’s Three Design Dimensions

To contextualize our co-design and comparison studies, we outline three key design dimensions of GenUI. In contrast to general guidelines for designing human-AI collaboration Horvitz (1999); Amershi et al. (2019); Wang et al. (2020) (e.g., when to invoke AI and recover from its error), the following dimensions are grounded in the end-to-end process of interacting with GenUI, specifically addressing what input users should provide and how, whether the workflow should delve into one design or explore breadths first, and which fidelity is most appropriate as output.

3.1 Unstructured vs. Structured Input

HCI research has long contemplated the benefits and trade-offs between unstructured and structured input, from early work of a structural analysis on input devices Card et al. (1991) and subtasks Foley et al. (1980) to research that envisaged unstructured input manifested as natural language interaction Bolt (1980). This unstructured, conversational approach has since become a hallmark of mixed-initiative human-AI collaboration Horvitz (1999) and recently gained significant traction owing to the advances of LMs. Unstructured prompting, commonly found in chat-based AI interfaces, works well for non-technical users, as it is highly expressive and requires no learning curve Mishra et al. (2025); Morris (2024). LMs can respond to prompts by producing numerous outputs instantly, sometimes leading to serendipitous discoveries Brade et al. (2023); Suh et al. (2024). However, crafting effective prompts to accurately capture creative intent is difficult Brade et al. (2023); Riche et al. (2025); Mishra et al. (2025), often involving laborious trial-and-error Brade et al. (2023) and a high cognitive load for non-experts who may lack the necessary vocabulary or domain knowledge Brade et al. (2023); Riche et al. (2025); Mishra et al. (2025); Morris (2024). This often leads to premature convergence on limited ideas (fixation), preventing users from exploring the vast design space in a structured way Suh et al. (2024).

3.2 Depth- vs. Breadth-First Workflow

There has been a long-standing discussion on the tension between going deep vs. broad. Shneiderman’s mantra advocated starting with an overview (breadth) than filter and zoom in to examine details (depth) Shneiderman (2003). Design literature found that divergent (breadth-first) and convergent (depth-first) activities are both key to the creative process and often occur in cycles Goldschmidt (2016). Among creativity support tools, exposing users to a breadth of design possibilities has been a hallmark of the workflow Dow et al. (2011), either by sampling the underlying parameter space to show design examples Terry and Mynatt (2002); Xiang ‘Anthony’ Chen et al. (2016), by clustering similar designs to provide an overview Chang et al. (2018); Petridis et al. (2022), or by formulating high-level design dimensions to support systemic exploration Suh et al. (2024); Ma et al. (2025). Specifically related to GenUI, prior work considered UI prototypes as a means for “traversing a design space” Lim et al. (2008) and argued for exploring many than just a few Tohidi et al. (2006); Buxton (2010). When generating a specific UI design, the use of intermediate representation (e.g., editable UI components Leung et al. (2025); Yuan et al. (2025)) is another way of allowing designers to explore and control variations before converging to a specific version.

3.3 High- vs. Low-Fidelity Generation

Literature has shown that both low and high-fidelity prototypes have distinct roles and benefits depending heavily on the specific stage and goal of design. Low-fidelity prototypes, characterized by limited functionality and interaction, are valuable in the early stages for quickly generating and exploring a breadth of ideas Buxton (2010); Greenberg and Buxton (2008); Landay and Myers (1995); Rudd et al. (1996). They allow designers to focus on high-level structure and behavior, facilitating “getting the right design” and avoiding premature fixation on details that can stifle creativity Greenberg and Buxton (2008); Landay and Myers (1995); Rudd et al. (1996); Walker et al. (2002). Conversely, high-fidelity prototypes offer full interactivity and a realistic appearance, making them suitable for later stages to “get the design right” Virzi et al. (1996); Rudd et al. (1996). Premature application of high-fidelity methods can be harmful by quashing innovative ideas, misdirecting focus to minor issues, or failing to account for cultural adoption over time Greenberg and Buxton (2008).

To summarize, while these three dimensions are central to the emerging GenUI landscape, above-mentioned literature has largely examined them in isolation. We address this gap by systematically comparing the design trade-offs between these dimensions as manifested in two GenUI tools.

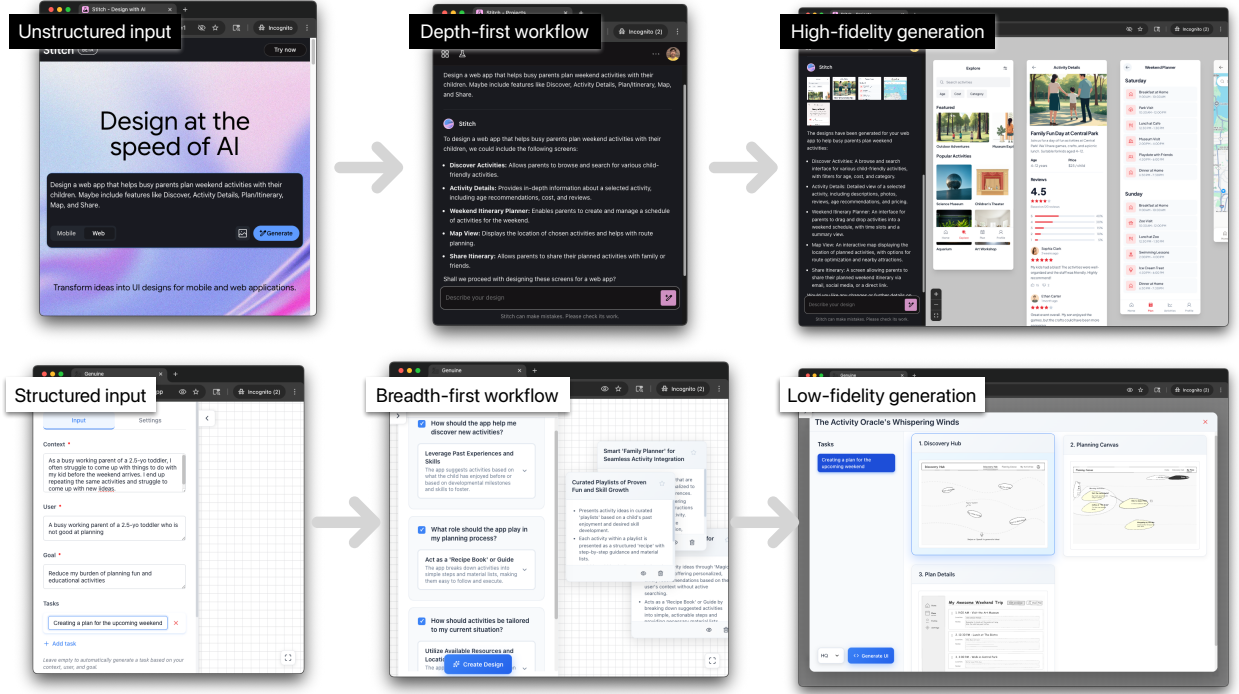


Figure 2: Comparison of two GenUI designs. Top: based on Stitch Google (2025b), starts with an unstructured prompt and delve depth-first into the generation of a high-fidelity UI mockup, whereas Bottom: our contrastive design, starts with a structured input and a breadth-first exploration of the design space, and then generates low-fidelity UI mockups.

4 A Tale of Two GenUI Designs

In this section, we describe the two designs of GenUI tool we aim to study: **STITCH** that employs an unstructured input and a depth-first workflow to generate high-fidelity UI mockups, and **MINDSPAN**—the contrastive one we created (implementation details in §5), which uses a structured input and a breadth-first workflow to generate low-fidelity UI mockups.

In the scope of this work, we consider GenUI’s target users as stakeholders involved in the early design ideation stage, which often consists of both PM and UXD (Chamberlain et al. (2006); Melegati et al. (2024)). To construct our narrative, we will follow Larry, a product manager (PM) with a background in design, as he uses these two different GenUI tools (dubbed **STITCH** and **MINDSPAN**) to explore ideas for features in a new software product—a Web app that helps busy parents plan weekend activities with their children. Below we will juxtapose and contrast how Larry performs each key step differently using the two GenUI tools.

4.1 Unstructured vs. Structured Input

STITCH As shown in Figure 2, after Larry starts a new project, he writes a single plain-English prompt just like how he uses other conversational large models: “*Design a web app that helps busy parents plan weekend activities with their children*”. Then he feels like adding a little more details, so he continues typing “... *Maybe include features like Discover, Activity Details, Plan/Itinerary, Map, and Share*”.

MINDSPAN In contrast, Larry needs to fill in a structured input form, including the four required CUI-based input fields—context, user, goal, and tasks, where the default placeholder text explains what each field is for. Larry is not prepared to have all this information ready, so he pauses to think about each field, *e.g.*, the goal of the parent using this app, which should go beyond simply keeping their children busy during the weekend. Larry considers a few possible goals, *e.g.*, to have fun, to stay connected as a family, and to educate the children. He decides that he wants to explore a combination of fun and educational activities and fills in the goal accordingly. Then, Larry also needs to fill in the tasks—what the user needs to do to achieve the goal, which forces him to think a little more concretely. Larry believes that the main task should be creating a plan for the upcoming weekend and he creates that one task accordingly.

4.2 Depth- vs. Breadth-First Workflow

STITCH As shown in Figure 2, after entering the prompt, Larry then hits a button that returns a list of screen descriptions in text and asks if he wants to proceed and generate those screens for an app mockup. Glancing at the bullets and seeing that the descriptions do include what he mentioned earlier in the prompt (e.g., Discovery Activities, Activities Details), Larry gives **STITCH** the go-ahead to generate the UI mockups.

MINDSPAN After entering the CUJ information, **MINDSPAN** first creates a design space that consists of multiple dimensions, each phrased as a question calling for Larry to ponder a key design decision, e.g., “What role should the app play in my planning process?” Larry can see and select options to respond to each dimension’s question: each option leads with a short summative phrase (e.g., “Act as a ‘Recipe Book’ or Guide”), followed by a brief explanatory sentence. Larry can also uncheck a dimension if it feels less relevant. **MINDSPAN** persistently shows the design space as a panel on the left side of the screen, allowing Larry to revisit it later to explore other different designs.

Next, Larry clicks the “Create Design” button to instantiate a specific idea based on the selected options in the design space. **MINDSPAN** shows a design as a stick-note-like card that summarizes the high-level concept as bullet points, placed on a canvas that supports direct manipulation, e.g., separating designs into groups, or arranging them close to one another for comparison. Larry can also star a design card to mark it as a favorite, or delete it.

4.3 High- vs. Low-Fidelity UI Generation

STITCH As shown in Figure 2, **STITCH** generates high-fidelity screen mockups with polished typography, spacing, and components. Larry clicks into each generated screens and begins editing in place using the customization panel (e.g., changing colors) or back to the conversational panel via interactive chat—for example: “*On Discover, add filter chips for ages 2–4 / 5–7 / 8–10 / 11–13; add Free and Indoor toggles*” and “*On Map, add route preview for the current itinerary*”.

MINDSPAN Larry selects one design card that interests him and opens its UI view, which first shows how the task—specified earlier in the structured input—corresponds to a flow of screens, each described in a one-liner text (no mockups generated yet). Intrigued by the descriptions, Larry proceeds to generate UI codes for the screens using a default fast mode, which enables a faster (albeit rougher) “preview” version of the design³. As shown in Figure ??, the generated UI codes render each screen as a wireframe-like low-fidelity prototype. Larry opens the full view of a screen, navigates to the previous or next one in the task flow, and reads about how a user would interact with each screen to perform the task. Then, he closes the UI view and goes back to the deck of design cards.

5 Implementation Overview

We describe a high-level overview (Figure 3) of **MINDSPAN**’s implementation, while leaving low-level details (e.g., step-by-step prompts in **MINDSPAN**) in the Appendix A.

We implemented the front-end of **MINDSPAN** as a Web app using Vite Vite (2025) with React React (2025). The back-end consists of prompting a series of Gemini models Google (2025a): 2.5-pro, 2.5-flash, and 2.5-flash-lite. However, the implementation was not tied to specific models; rather, the choice of models for different steps was mainly a trade-off between performance and speed. Therefore, below we do not differentiate which model we used for which step.

First, to create a design space, we employ a chain of two steps. Given all the provided input (e.g., context and CUJ information), we prompt the model to *brainstorm n different ideas*. We chose $n = 15$ to strike a balance between sufficient diversity and preventing additional latency. Then, we prompt the model to analyze the ideas to *identify common design dimensions* and create multiple options to choose from for each dimension.

To create a specific design, we prompt the model to implement the selected options in the design space and *generate n distinct high-level design concepts*, from which we *randomly select one* to show to the user. We chose $n = 5$ for the similar reason of balancing diversity and latency. Each model-generated design consists of a concise description and a detailed description framed in a way that a UX designer can understand and implement the concept into specific UI screens. Next, we prompt the model to build on the generated design and create textual descriptions of each UI screen. Given that there might be multiple tasks the UI screens need to support, we first prompt the model to generate *task-wise screen descriptions*—that is, each task has its own sequences of screens, and then combine them into a *unified sequence that support all tasks*.

³A high-quality (HQ) but slower mode is also available

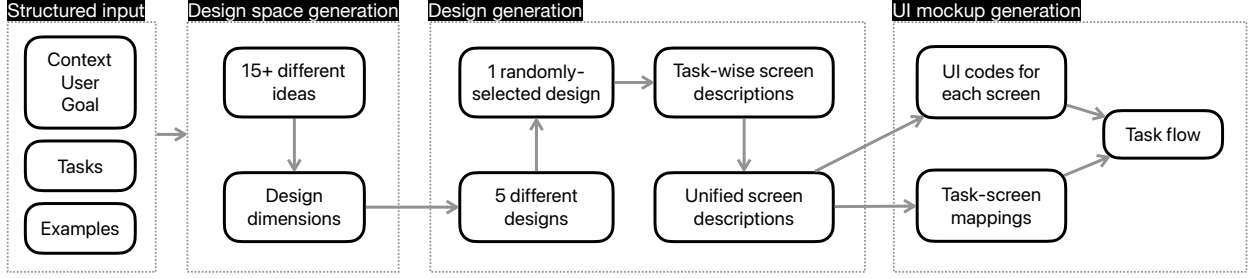


Figure 3: **Block diagram of MINDSPAN’s implementation.**

Then, we prompt the model to generate *task-screen mapping*—to look back at each individual task and identify which screen(s) support that task, creating a mapping between each task and the corresponding screens.

To generate *UI code for a screen*, we prompt the model with the screen’s textual description to generate SVG code that renders a low-fidelity wireframe-like mockup. We choose SVG over conventional HTML because its shape-based rendering naturally aligns with the look and feel of wireframes, while HTML’s layout system tends to resemble polished, high-fidelity screens. Finally, to generate a *task flow*, we use the task-screen mapping to retrieve its screens’ UI codes. For each screen involved in this task, we prompt the model to examine its key interaction (generated as part of the screen description) in the context of the generated UI codes and to describe the user interaction—how a target user might interact with the screen. We present the user interaction description alongside the rendering of each screen’s UI code, thus presenting how the user progresses in the task on this screen.

6 Comparison Study

We conducted a series of hands-on sessions to study how the two designs of GenUI tool support users’ 0-to-1 exploratory design tasks, focused on juxtaposing their benefits and trade-offs.

We hypothesize that—

- **Input:** Compared to unstructured prompt, users will benefit more from structured input in breaking down a high-level 0-to-1 design task into more specific facets, although at the cost of more cognitive load and time spent.
- **Workflow:** Compared to the depth-first way of “getting a design right”, users would find a breadth-first approach more appropriate for the 0-to-1 exploration although the large number of designs are challenging to process and keep track of.
- **Output:** While high fidelity generation is more realistic and polished, it also draws more attention to the low-level details (e.g., colors and fonts); in contrast, low fidelity allows users to focus more on the high-level concepts and its lighter information load makes it more agile for further development.

6.1 Participants

We recruited participants in a large North America software company via three different internal communication channels. There were two inclusion criteria: (i) the role was either UXD or PM (whose work is most often related to 0-to-1 design Chen et al. (2025)) and (ii) the response to the screening question “How is UX design related to your work?” was either “My main role is in UX design” or “UX design is a significant part of my work, though not my main role”. There were no exclusion criteria. Note that we followed prior work’s approach Chen et al. (2025) and included both PM and UXD, *not* to compare these roles, but to ensure that we could gather insights from both who are often involved in the early phase of design ideation. At the end, we recruited 24 participants, 12 UXDs and 12 PMs, from 15 different product teams. Among them, there were 13 male, 8 female, 2 non-binary, and 1 prefer-not-to-say; aged 30 to 51, with professional experiences ranging from 4 to 20 years. Participants were randomly assigned to one of four study sessions that took place between August 20 and 25, 2025. Our study has been approved by the IRB of our institute and followed participants’ company’s policy on engaging in research with human subjects. Participants were paid a \$40 gift card for their time.

6.2 Design, Tasks & Procedure

We employed a within-subject design. The independent variables was GenUI *Tool*: Tool A (STITCH Google (2025b)—a state-of-the-art GenUI tool representing the existing paradigm) vs. Tool B (MINDSPAN—our own prototype). A participant used each of these tools to conduct a different exploratory 0-to-1 design task and was randomly assigned to one of the following study sessions where the order of the tools and the task-to-tool assignment was counterbalanced:

- Session #1: Task 1 (MINDSPAN) → Task 2 (STITCH)
- Session #2: Task 1 (STITCH) → Task 2 (MINDSPAN)
- Session #3: Task 2 (STITCH) → Task 1 (MINDSPAN)
- Session #4: Task 2 (MINDSPAN) → Task 1 (STITCH)

We described each task to the participants in the following format (the two tasks only differed in the “new application” part):

Your team is exploring a <new application>.

Objective: Use the assigned GenUI tool to explore design ideas and propose one promising concept.

Suggestion: start by fleshing out the task description before using the GenUI tool (you can use other tools).

For Task 1’s new application, we adopted and revised the one used in Chen et al. (2025)—a mobile app that helps knowledge workers manage work-related stress. Task 2’s application was a desktop app that helps parents plan weekend activities with their children. We intended for the two tasks to be complementary: Task 1 was a mobile app, focused on fostering personal health during day-to-day work; while Task 2 was a desktop app, focused on maintaining family connection on the weekends.

We started each study, after seeking informed consent from participants, with an introduction of GenUI and the purpose of this study, *i.e.*, comparing two GenUI designs. Next, participants proceeded to the trials of using both Tool A and B. Each tool started with a video tutorial demonstrating their basic interactive features and functionalities. Next, we briefed participants with the task description, followed by information to access each tool. Then, participants spent 20 minutes on the design exploration tasks using the tool. We allowed them to use other tools if needed. After each trial, participants fill out a questionnaire to rate their experiences on the input, workflow, and output of the tool. Each session took place virtually on Google Meet and lasted about one hour.

Within two workdays after each design session, we conducted a 1:1 semi-structured interview with each participant to discuss their experiences and compare the three design differences between the two GenUI tools: structured vs. unstructured input, breadth- vs. depth-first workflow, and low- vs. high-fidelity generation.

6.3 Data Collection & Analysis

Immediately after each GenUI tool’s design session, we collected participants’ responses to 7-point Likert-scaled questions (Table 2), corresponding to the three key aspects of comparison: input, workflow, and output. We conducted Wilcoxon signed rank tests to understand whether there was any statistically significant difference in how they rated each GenUI tool. Further, we conducted Mann Whitney U tests to see whether PM and UXD participants responded differently to each question regarding either GenUI tool.

We collected qualitative data through the post-session interviews via taking notes of our discussions with participants. We then employed a semi-structured thematic analysis Terry et al. (2017). The three key aspects formed a preliminary thematic structure to group the initial data points. Within each thematic group, we then conducted further analyses, using *benefits*, *trade-offs*, and *suggestions* as seed categories to guide a two-pass coding of participants’ feedback. The first author conducted the first pass of coding, which was then reviewed by one other author. Disagreements between the authors were resolved through discussion, ensuring a comprehensive and consistent interpretation of the data.

7 Findings: Usage Patterns of GenUI Tools

To reveal usage patterns on the two GenUI tools, we analyzed screen recordings of participants’ design sessions, which show how they exhibited different behaviors across various phases from initial input to mockup generation and to iteration (Table 1). Further analyses of interaction logs on MINDSPAN examined its usage through a more detailed lens (Figure 4), showing how participants’ time was distributed over different phases, how they balanced “exploring the breadth” and “going deep”, and how they switched between these two activities.

7.1 Findings from Screen Recording Analyses

Table 1: A comparison of typical usage patterns of the two GenUI tools, based on a thematic analysis of screen recordings.

	STITCH	MINDSPAN
Initial input	Heavy editing, going through the prompt back-and-forth multiple times	Entering each input field once, with minimal or no edits
↓		
Pre-mockup-generation	Brief, single-step, reviewing an AI-generated list of screen descriptions	Elaborate, multi-step, selecting multiple options in the design space to generate and skim one or more specific designs
↓		
Post-mockup-generation	Surface-level browsing of mockups, occasionally delving into specific components on a screen	Only viewing 1-2 designs’ mockups, following the task flow to go through each screen
↓		
Mockup iteration	Frequent and low-level, across multiple places: follow-up prompts, customization panel, and comparing screens between iterations	Few and mixed-level, either via adding to or modifying initial input or revisiting the design space to select different options

As shown in Table 1, we compare participants’ usage patterns of the two GenUI tools across four common stages of GenUI interaction: initial input, pre-mockup-generation, post-mockup-generation, and mockup iteration.

During initial input, one noticeable contrast is that **STITCH** users tended to edit their prompts heavily, going through the prompt back-and-forth multiple times. For example, UXD3 first briefly described an idea, then added stylistic descriptions (color schemes and layout); UXD9 edited the prompt twice—first adding layout details then navigation features. In contrast, with **MINDSPAN**’s structured input, users entered each field once, with minimal or no edits. Such differences suggest that **MINDSPAN** users might have thought through their response to each field before entering it, whereas **STITCH** users might have treated the unstructured prompt as a working draft that resulted in the need to refine it iteratively.

Next, at the pre-mockup-generation stage, **STITCH** users performed a brief review of an AI-generated list of screen descriptions. In contrast, **MINDSPAN** users had to go through a more elaborate, multi-step process, going back-and-forth to select multiple options presented to them in the design space, before creating a specific design. Some changed their design option selections multiple times and generated different designs, whereas others would move on to a design first and return later to the design space to explore other options (described below in the iteration phase). For participants who created multiple designs at once, they would skim through all of them first (PM3, UXD2), although some paused mid-way to linger more on one design (UXD8) or to open its UI view for more details (UXD4).

After the tool generated UI mockups, both **STITCH** and **MINDSPAN** users browsed through the screens at a surface level (*i.e.*, a quick scan). The main difference is that **STITCH** users occasionally selected and zoomed into specific components on a screen whereas **MINDSPAN** did not provide such “deep-dive” features. **MINDSPAN** users only viewed one or two selected designs’ generated mockups, following the task flow to go through each screen.

Then, in terms of iteration, **STITCH** users tended to make frequent and low-level changes across multiple places: follow-up prompts, customization panel (*e.g.*, adjusting fonts and colors), and comparing screens between iterations. For example, PM7 added layout requirements (grid) as a follow-up prompt and UXD1 adjusted both fonts and colors on the customization panel. We did notice the following participants attempted to explore high-level changes as new feature requirements, including advanced filtering function (PM6), dashboard-style design (UXD4), and map view (UXD9). However, their attempts were unsuccessful as the tool either ignored them or failed to generate relevant results. Participants seemed to have a tendency to be attached to the first design and our analysis found no one started over with a new prompt to pursue a different design. In contrast, **MINDSPAN** users made few and mixed-level changes mainly in two ways: some went back to the input fields and add either stylistic (UXD2, 4) or functional requirements (UXD3, 9) and some changed their selections in the design space (UXD2, 8, 9).

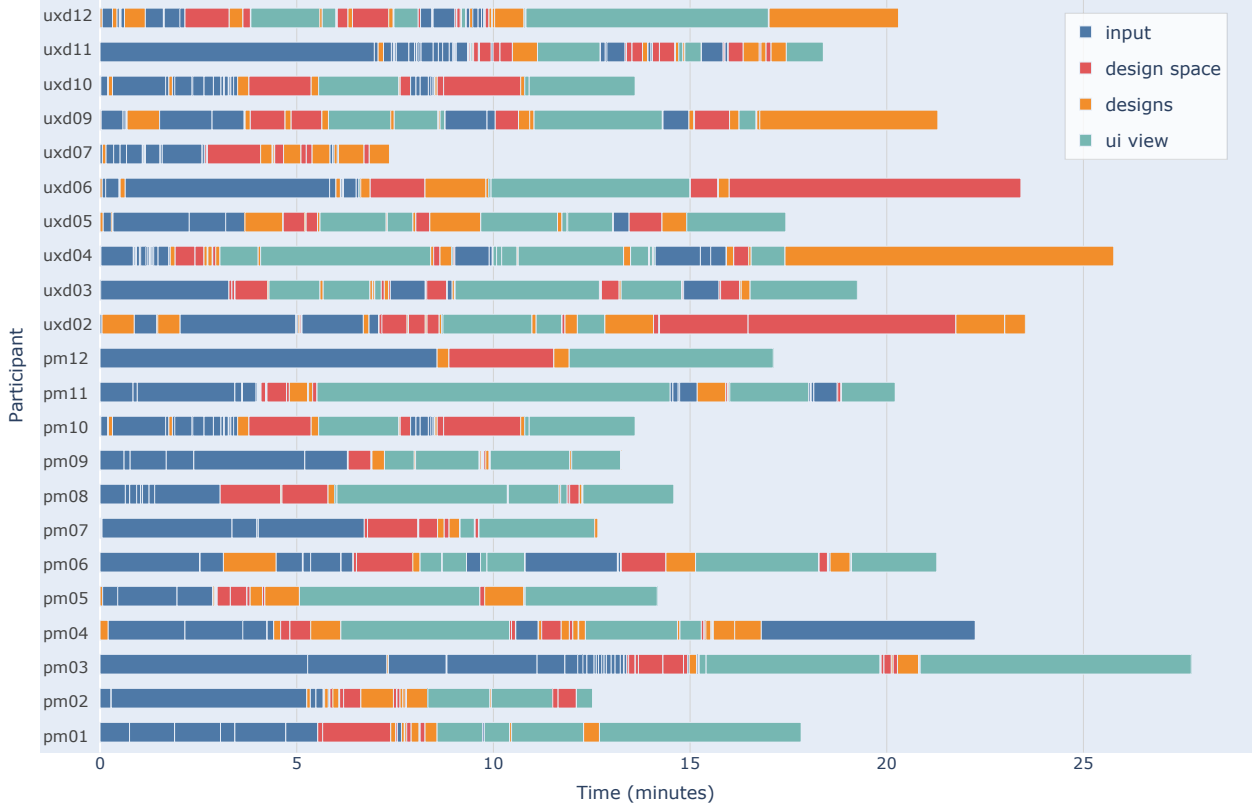


Figure 4: **A time-series visualization of participants’ interaction with MINDSPAN.** Participants spent various amounts of time on the four different parts/steps of MINDSPAN over the courses of their design session ($\mu = 18.1$ min): a significant amount of time on the structured input ($\mu = 5.9$ min), followed by switching back and forth between exploring the breadth (design space, designs: $\mu = 5.5$ min) and going deep (UI view: $\mu = 6.7$ min).

7.2 Findings from MINDSPAN’s Interaction Log Analyses

To complement qualitative insights based on screen recordings, we further collected and analyzed interaction logs on MINDSPAN from all but two participants⁴. By associating each logged event with a specific major component of the tool, we were able to break down the timelines of each participant’s interaction with MINDSPAN (Figure 4). Note that our logging mechanism did not employ a hard cut-off and, as a result, the lengths of participants’ timelines varied: Some were much shorter likely because they started with using other tools (e.g., Gemini) to prepare for the input, which we did not log; a few were much longer likely because they stayed in the tool or revisited it (e.g., for answering survey questions) even after the session was concluded.

Many participants spent quite a portion of their time entering the structured input. Further analyses revealed that participants spent an average of 5.9 minutes ($SD = 2.9$ minutes) on input, which is about one third of their entire session (average is 18.1 minutes). Noticeably, some participants (e.g., PM7, PM12, and UXD11) spent over half of their sessions on input. Such behaviors suggest that entering structural input is a non-trivial, or even time-consuming step, which we discussed later in the interview findings.

The rest of participants’ timelines were divided between interacting with three components: the design space panel, the designs (presented as design cards), and each design’s UI view. By and large, we can consider interacting with design space and designs as “exploring the breadth” whereas delving into a design’s UI view as “going deep”. Further analyses show that participants spent an average of 5.5 minutes ($SD = 3.5$) exploring the breadth and an average of 6.7 minutes ($SD = 3.1$) going deep. Some participants were more breadth-focused, spending over $2\times$ the amount of time on design space and designs as on UI view (UXD2, 6, 7) whereas quite a few were more depth-focused, spending over $2\times$ the amount of time on UI view as on design space and designs (PM1, 3, 8, 9, 11; UXD3). To understand how participants switched between breadth and depth activities, we counted the segments of UI view interaction. Intuitively,

⁴Due to technical issues, they were unable to access and submit their browser’s log file.

Table 2: **Results of the post-session survey.** We asked each participant to rate (on a 1 to 7 Likert-scale) their agreement or disagreement with each statement regarding both GenUI designs. The heatmap visualizes the distributions of their ratings. We then conducted a Wilcoxon signed-rank test on participants’ ratings between the two GenUI designs, and Mann–Whitney U tests to further understand how UXD and PM rated each GenUI design differently.

	Survey Question	VibeDive								MindSpan								VibeDive vs. MindSpan		PM vs. UXD: VibeDive		PM vs. UXD: MindSpan	
		1	2	3	4	5	6	7	Mdn	1	2	3	4	5	6	7	Mdn	W	p	U	p	U	p
Input	The input method helped me clearly articulate my design requirements.	1	4	1	4	4	6	4	5.0	1	3	5	1	5	7	2	5.0	85.0	0.70	36.5	0.04	56.0	0.36
	I felt confident that the tool understood what I wanted to design.	0	1	3	6	6	5	3	5.0	0	6	1	4	8	4	1	5.0	65.0	0.24	48.5	0.17	72.0	1.00
	It was easy to provide all the necessary details for my design task.	1	2	6	0	3	6	6	5.5	1	5	5	2	5	4	2	4.0	45.5	0.15	47.5	0.16	43.5	0.10
	The input guided me to consider aspects of the design I might have otherwise missed.	1	6	4	3	3	4	3	4.0	3	2	1	1	7	8	2	5.0	84.0	0.28	51.5	0.24	69.5	0.91
Workflow	The workflow encouraged me to compare different design alternatives.	3	7	2	1	4	5	2	3.5	1	1	1	3	6	8	4	5.5	28.0	< .01	71.0	0.98	57.5	0.40
	I was able to explore a wide range of possible design directions before deciding on one.	3	10	6	1	1	2	1	2.0	1	3	6	2	5	5	2	4.5	19.0	< .01	64.5	0.67	61.0	0.54
	The tool helped me consider design ideas I would not have thought of on my own.	2	6	1	7	4	3	1	4.0	0	3	2	0	7	5	7	5.5	25.0	< .01	63.5	0.64	60.5	0.51
	The tool gave me a good understanding of the overall design space.	2	3	2	4	4	7	2	5.0	3	8	2	2	4	4	1	3.0	44.0	0.07	47.5	0.16	62.5	0.59
Output	The fidelity level matched my needs for the stage of the design process.	0	2	4	2	5	5	6	5.0	7	4	6	2	2	2	1	3.0	24.5	< .01	45.0	0.12	64.0	0.66
	The mockups allowed me to focus on high-level concepts rather than fine details.	0	2	5	0	7	5	5	5.0	2	2	2	4	7	4	3	5.0	75.0	0.27	55.5	0.34	71.0	0.98
	I can use its generated mockups to communicate my ideas to others.	1	0	0	1	6	6	10	6.0	4	3	3	5	4	2	3	4.0	4.0	< .01	52.5	0.25	66.5	0.77

having multiple UI view segments means the user frequently goes deep and then switches back to breadth. On average, participants had 3.0 UI view segments ($SD = 1.4$), *i.e.*, switching between breadth and depth three times.

8 Findings

For each of the three design aspects, we first summarize quantitative results of survey questions and then discuss recurring themes distilled from qualitative interview data. For each key finding, we highlighted whether it was shared between PM and UXD, or more pertinent to a specific role.

Overview of PM and UXD’s Differences in Findings: Across the three design dimensions, most findings were shared between PM and UXD participants, except for the following differences: (i) More PM participants reported struggling with filling out the structured input at the beginning of design exploration; (ii) PM seemed to appreciate the speed and efficiency of the depth-first workflow more while UXD felt limited by its focus on the details of look and feel; (iii) More PM were drawn to a high-fidelity output as it seems to have brought them closer and faster to a product-ready state.

8.1 Unstructured vs. Structured Input

We asked four input-related questions during the post-session survey (Table 2). Statistical analyses show that, for each of the questions, there is no significant difference between participants’ ratings of the two GenUI input designs. As analyzed below with qualitative data, the lack of difference is likely because either way of input has its own unique strengths and struggles and neither offers a better solution than the other to express one’s UX design needs.

Participants appreciated how the structure surfaced key UX-specific facets and “*clearly lays out the things that will control the output*” (PM11), making them think through the problem harder (UXD3-5, 9; PM6, 11). The structure is more than a template (UXD5, PM3); its specificity served as a scaffold for participants to deepen their thoughts:

Initially did not differentiate tasks from goals; having this structure forces me to think at a more granular level. (UXD9)

Tasks helped to bridge the design to flows and allow me to focus on one part of the experience and jump to others—serving as a structure to keep me aware of which parts of the user journey are in the design. (UXD8)

What is surprising is the overwhelming complaints from two thirds of the PM on **the difficulty and their unpreparedness of starting exploration with structured input, with particularly more struggle to fill out “tasks”**. Specifically, 8 different PM’s feedback echoed such complaints, while only 2 UXD shared a similar sentiment. Foremost, from a UI point of view, participants felt the “form filling” experience too heavy (PM6), overwhelming (PM2), or even intimidating (UXD1). Participants felt such a structure “*harder to get started (with)*” (PM8) for design exploration—“*I haven’t soaked in the idea; do not want to think about these things before visualizing something*” (PM1). The fundamental problem is that, to some participants, asking for specific information upfront felt premature and lacking openness, “*forcing*” (PM3) them to think through many aspects before they were ready (PM1, 3, 6). Further, some participants also had initial ideas that “*don’t fit any of those boxes*” (UXD5) and they did not know where to input those kinds of information (PM7).

In contrast, both PM ($n = 8$) and UXD ($n = 11$) found that unstructured input, though imperfect, more appealing because of a more open format and a lower barrier of entry. Participants did point out many challenges of unstructured prompting that echoed findings from prior work Arawjo et al. (2024); Zamfirescu-Pereira et al. (2023); Chen et al. (2023); Subramonyam et al. (2024a); Mishra et al. (2025), e.g., difficulty of formulating effective prompts (PM2, 11; UXD2, 5, 6, 8, 11, 12) and the uncertainty of not knowing what or how much to specify in the prompt (PM3; UXD1, 3, 6). Nonetheless, they preferred this way of input upfront for its openness (PM2, 10; UXD4, 10, 11) and flexibility (PM4, 7, 10; UXD4, 9) that could help them get started easily (PM1, 8, 11).

8.2 Depth- vs. Breadth-First Workflow

We asked four questions on how each GenUI tool’s workflow supported the assigned design exploration task, based on the rationale that diverging to alternatives is key to prevent fixation at the early stage Design Council (nd); Dow et al. (2011); Goldschmidt (2016). As shown in Table 2, participants rated MINDSPAN higher on the following statements: (i) “*The workflow encouraged me to compare different design alternatives*” ($W = 28.0, p < .01$); (ii) “*I was able to explore a wide range of possible design directions before deciding on one*” ($W = 19.0, p < .01$), and; (iii) “*The tool helped me consider design ideas I would not have thought of on my own*” ($W = 25.0, p < .01$). These ratings indicate that participants did benefit from using the breadth-first MINDSPAN for design exploration. Interestingly, we did not find statistically significant difference in the fourth question “*The tool gave me a good understanding of the overall design space*”, nor was there differences in how PM and UXD rated each GenUI tool on any of the four questions.

Overall, both PM and UXD participants recognized the benefits of the breadth-first workflow (UXD6, 10, 12; PM2, 6, 8, 10, 12). Dissecting the design task into different dimensions in the design space allowed them to “*think about combinations of things*” (UXD4) and to “*mix and match different ways of responding to design needs*” (UXD9). Such an early exposure to a “*wider field*” (PM10) is important for the following reasons.

UX design is highly multi-faceted and an exploration often requires considerations of users’ persona, product metaphor, mental models of task flow, among many other things. **An equal number ($n = 5$) of PM and UXD mentioned how a breadth-first workflow challenged their (implicit) assumptions on these UX design facets**, pushing them to research (PM2) and test hypotheses (PM3), seeing the “*polarity*” (UXD4) of ideas. As a result, the breadth-first workflow allowed participants to explore ideas they would not have thought of (PM9, 12; UXD3, 5), e.g., “*different factors that made experiences different from one another*” (UXD2), as well as user journeys “*different than what I would’ve thought of*” (PM8) and that they “*didn’t think through when getting started*” (UXD8).

.. the (MINDSPAN) app more full ... [It] gave you these spectrums [of] target directions, helping you think through the breadth of what the app could be, felt more complete. (UXD4)

... the tool made me think about the aspect of AI’s persona—would not have come to if I had used the other tool. (PM3)

Participants—both PM ($n = 2$) and UXD ($n = 2$)—also treated a breadth-first workflow as a thoughtful, analytical, and unhurried engagement with the design problem. It allowed them to take a slower (UXD1, 3) pace, “*go through a more granular process*” (PM8) where the questions in the design space helped them “*step back and look at the big picture*” (UXD3).

... slower pace [with] different directions of problem solving: stop, pause—we got the basic here; these are the things I will think about before interviewing users. (UXD3)

Like a sounding board: instead of saying “what you have is perfect, let’s design it,” it asks “what if this” and “... that”. (PM12)

In terms of trade-offs, participants noted how the breadth-first workflow took longer (UXD11), more steps (UXD1), and a lot of “pogo stick like” (UXD9) back-and-forth (PM7) before they got to see some UI mock-ups. To navigate the design space, participants mentioned the sheer number of options, having to manage the possibilities of 3 questions $\times$ 4 options each (PM1, 5; UXD10). The sticky-note-sized, bullet-pointed text descriptions of each design idea, felt “too busy” (UXD10) and “too technical ... text-heavy” (PM2, 6) and the freeform placement lacked organization (PM6), hard to navigate (PM6, UXD5). A lack of visual (PM2, UXD11) created a barrier for design exploration: UXD6 mentioned how they were used to seeing design as mockups rather than the textual design cards and UXD11 felt an exploration was inadequate without actually generating the mockups.

Indeed, both PM and UXD participants’ main struggle seemed to be the challenge of effectively and efficiently previewing UX design ideas at a glance, as mentioned above by 5 PM and 6 UXD. While some other generated contents can be browsed at a glance of their snapshots (e.g., sentences Reza et al. (2024), images Evirgen and Chen (2022) and 3D models Matejka et al. (2018)), it is unclear how to present a single preview of a 0-to-1 UX design. Each UX design often consists of multiple screens to support multiple task flows: presenting all of them in one visual storyboard is too overloaded (and time-consuming to generate); only showing one screen is feasible but cannot fully represent the entire design. We discuss later in §9 possible solutions to address this challenge.

In comparison, **participants (4 PM, 1 UXD) felt the depth-first workflow excels at speed and efficiency.** Participants liked how it produced a design fast (PM7, UXD1), which felt like making quick progress (PM5, 10), getting “closer to done faster” (PM10). Participants pointed out that such a “PM mindset” (PM2, 5, 10) played a role in their preference of depth-first workflow, which allowed them to prioritize making progress on fewer design options over researching on broader alternatives (PM5, 10). **However, both PM ($n = 5$) and UXD ($n = 8$) pointed out that the depth-first workflow remains limited in supporting design exploration—**“not as blue sky and wildly creative as breadth-first approach” (PM10) and “didn’t really help me explore” (PM9). Foremost, delving into a design right away was “too focused on one solution” (UXD11) while it remains “unclear what to do next with the generation” (UXD3).

It talked back to me what I was thinking and gave me one option, asking if I wanted to change anything, as opposed to suggesting other options. I am still limited to what I was thinking about, not outside of the box. (PM12)

Further, UXD participants, in particular, mentioned their tendency to focus on “look and feel” (UXD4) and “tactical details” (UXD3), e.g., colors and fonts, when given the one design generated upfront. The depth-first approach seemed to have “framed my mindset into focusing on visual refinement” (UXD4) and “made me think we could come up with something ‘done’ and just finetune it” (UXD3) even though they realized they should “instead focus on flow and whether the right information is placed” (UXD10). Such feedback was also related to the fidelity aspect, which we discuss below.

8.3 High- vs. Low-Fidelity UI Generation

As shown in Table 2, participants rated the high-fidelity approach higher on the following two questions: (i) “The fidelity level matched my needs for the stage of the design process” ($W = 24.5, p < .01$); and (ii) “I can use its generated mockups to communicate my ideas to others.” ($W = 4.0, p < .01$). As discussed below, such preferences were largely framed by participants’ professional practices. Participants gave a median rating of 5 of both tools (no significant difference) for the question “The mockups allowed me to focus on high-level concepts rather than fine details”. To many participants, low fidelity’s lack of fine details was also an impediment in conveying the design concepts. We did not find any significant differences between PM’s and UXD’s responses to the fidelity-related questions.

Echoing prior work Virzi (1989); Landay and Myers (1995); Buxton (2010); Greenberg and Buxton (2008), both PM ($n=4$) and UXD ($n=5$) participants did recognize many positive aspects of low fidelity, e.g., its appropriateness for 0-to-1 design exploration (PM11; UXD8, 10), its focus on interactions (PM3, UXD12) while deferring judgments on details (UXD2, 10), and the freedom and flexibility knowing that it is not a finished product (PM9). Participants also showed a clear understanding of high fidelity’s drawbacks, e.g., its details become distractions from the core design (PM3, 4), drawing too much of people’s attention to minutiae (PM9; UXD3, 10).

Despite the many reasons to favor low fidelity for design exploration, more PM ($n = 7$) than UXD ($n = 3$) participants still mentioned a preference of high fidelity. Besides the obvious reason that high fidelity being more visually and interactively appealing (PM4, 10, 11; UXD9), participants also mentioned their desire for getting to a high-fidelity end-state fast (PM3, 5, 6, 11; UXD5, 10) where advanced AI can help them skip the previous need to convert from low to high fidelity (PM2, 11; UXD5). Such mindsets echoed the earlier finding of how some participants preferred depth-first workflow because of its speed and efficiency.

Both PM ($n = 6$) and UXD ($n = 8$) indicated how high fidelity better fits participants’ professional UX practices whereas low fidelity was considered unwieldy. First, participants considered high fidelity more compatible with UX work, providing a *“good version to start working off of”* (PM11) and better for hand-off to downstream tasks (UXD4). Contrary to our hypothesis that low fidelity’s light information load makes its further development more agile, participants considered low fidelity *“slows down the design process”* (PM3) and *“harder to build off (of)”* (UXD10). Such judgments seemed to have stemmed from participants’ tendency to measure the distance between the low-fidelity and the eventual state, leading to the mindset that higher fidelity equals more progress and closer to the finish line.

Second, working with high-fidelity mockups felt familiar to state-of-the-art professional tools, e.g., Figma (UXD2, 5, 6), and closer to how they *“use design systems in work”* (UXD8); in contrast, low fidelity felt outdated, e.g., like the early wireframing tools (PM6, UXD10), and *“unlike real design systems”* (UXD5).

Finally, high fidelity serves better for other stakeholders, e.g., *“for engineers and power users to look at detailed and more engaging mock(ups)”* (PM3), for showing live demo (UXD11), and for getting feedback from others (UXD2). In contrast, low fidelity is *“hard to explain to a user”* (UXD1) and cannot be handed off due to its openness that causes ambiguity (PM5), thus requiring more UX work and resources (PM5, 8).

Such a “high-fidelity mindset”, combined with a high expectation built up through the breadth-first exploration, amplified participants’ disappointment when presented with low-fidelity mockups. Having developed a clear understanding of user journey and tasks (UXD8) and explored various designs in textual forms (PM12), participants felt seeing a low-fidelity output mismatched the ideas that had emerged in their mind (PM12)—a *“letdown”* (PM6) that *“made it hard to translate those (design ideas) to visuals”* (UXD8), *“undoing the effort spent on understanding the user and flow”* (UXD3). As one participant put it—

Fidelity too low; uninspiring compared to the quality of thoughts leading to it—cannot match the promise of the design ideas. (UXD9)

9 Implications for Design

In this section, we distinguish between findings observed in our study and design implications derived from them. The implications are not directly validated design solutions; rather, they are cautiously framed directions suggested by participants’ experiences and by our interpretation of the trade-offs surfaced in the study.

Finding: neither unstructured nor structured input was uniformly preferable at the outset of exploration. Our survey and interview findings (§8.1) indicate that, at the outset of design exploration, there is no definitive advantage to either unstructured or structured input. Participants’ suggestions often gravitated towards a hybrid, mixed-initiative Horvitz (1999), unstructured-to-structure approach, but our study did not directly evaluate such a hybrid design. Such a hybrid input dialog may be especially worth exploring for PM users, as we observed more PM struggled with structured input in the study. **Implication: GenUI may benefit from mechanisms that let users begin openly and add structure gradually.** Similar to how existing conversational LMs augment a vanilla prompt with suggestive chips or autocompletion, GenUI could *“suggest a more structured way to enter the open-ended prompt”* (UXD5). As users start typing a prompt, the tool might proactively map contents to specific fields (UXD1), or reactively let the user select an option to rephrase or refine the prompt with more structure (PM5, UXD5), where the tool could in turn suggest missing information (PM12). Alternatively, users can further defer formalizing the critical user journey, perhaps explore some designs first, and later build on their initial unstructured input with *“a guided conversation about users, goals, and tasks”* (UXD12). For users who prefer a “warm start”, these mechanisms may make it easier to react to something tangible before developing a structural understanding upfront.

Finding: breadth-first GenUI made previewing many UX designs a central challenge. Our findings suggest that the crux of breadth-first GenUI may be effectively and efficiently previewing UX designs, each of which itself often consists of multiple screens to support multiple tasks. **Implication: future GenUI systems could explore richer ways to summarize and compare alternatives.** Three possible directions emerged in the discussions with participants: (i) Start depth-first, generate a “seed design” as an anchor, and then extrapolate the breadth of other possibilities—*“give me something tangible, then try different variations to suggest ideas I hadn’t thought of—classic brainstorming”* (UXD3). (ii) Show design alternatives side-by-side via a competitive analysis Rosson and Carroll (2002) table: To realize this approach, a system would likely need to identify the critical components or interaction flow on each UI that are worth comparing, as well as highlight key differences, e.g., using a mix of modalities—textual summarization and visual annotations on the UIs. (iii) Generate a rapid serial visual presentation (RSVP) Potter (2018) of UI states—a brief animation of the interaction flow as the preview of a design idea. Ideally, in just a few seconds, a user could watch a few key frames and grasp how a task is performed using the generated screen mockups. Rather than rendering each screen

in full details, the RSVP could adopt an abbreviated version, *e.g.*, only showing the layout and using icons to signify Norman (2008) each component’s functionality.

Finding: participants often expected generated mockups to look closer to product-ready artifacts. With a “high-fidelity mindset”, potentially heightened by the increasing popularity of “vibe coding” Sarkar and Drosos (2025), participants appeared to expect polished, near-product-like mockups as the norm of AI-generated artifacts. **Implication: mixed-fidelity generation may be a useful design direction to explore.** Rather than treating fidelity as a binary choice, future GenUI systems could consider generating high-fidelity UI mockups by default while also enabling conversions to mixed-fidelity, which was defined as “high-fidelity on some dimensions and low-fidelity on others” McCurdy et al. (2006). High-fidelity by default may cater to PM who prefer to see a near-end-product version of the design up-front. Meanwhile, enabling mixed-fidelity would allow users to dial up or lower the overall fidelity based on their preference, as demonstrated in early work on a mobile UI design tool de Sá et al. (2008). Further, rather than considering fidelity as a global parameter, future systems could let users select a subset of the UI components to locally change their fidelity. For example, low fidelity might be enough for conventional UIs (*e.g.*, buttons and text boxes), but users might require more details to better understand the novel components, *e.g.*, a treasure map for parents and children to discover weekend activities. Future developments of GenUI tools could investigate how to decide which components should be rendered at what level of fidelity.

Implications for other AI-powered creativity support tools. First, many other design tasks are also defined by multiple facets (*e.g.*, form, function, and cost for furniture design). Thus a hybrid, mixed-initiative, unstructured-to-structured input may help users ease into exploration while staying aware of what important information they might consider. Second, like UX, various other designs cannot be easily previewed at a glance, *e.g.*, narrative media (films, games, novels), interior designs, and services. To address this, future systems could explore creating one design first and then extrapolating, comparing alternatives side-by-side, or using rapid serial visual presentations. Third, fidelity is a general notion in other design domains, *e.g.*, the spectrum of sketches, scaled models, and photorealistic rendering in architecture. This suggests that mixed-fidelity generation may be a useful technique for allowing professionals to adjust the kinds and levels of detail they focus on based on the stage of their design process.

10 Limitations, Future Work, & Conclusion

Our study was intentionally scoped to early-stage, individual design exploration. We focused on PM and UXD, whose responsibilities align closely with exploratory 0-to-1 design tasks, and recruited participants from a single large company following prior work Subramonyam et al. (2024b); Chen et al. (2025). This focus allowed us to recruit more participants per role and compare their experiences in depth, but it also bounds the generalizability of our findings. Participants came from 15 diverse teams, yet their shared organizational culture may have shaped their expectations of GenUI. Future work should therefore expand across companies and industries, as well as to other UX-related roles such as front-end SWE, UXR, and cross-functional collaborators.

Our experimental setup also emphasized exploration over the broader arc of UX work. We scoped tasks to 0-to-1 design contexts where GenUI has shown strong potential Chen et al. (2025), leaving open how our findings extend to more common n -to- $(n+1)$ iteration. Similarly, our 20-minute exploration sessions enabled participants to try two tasks within an hour, but may not reflect longer-term use in which ideas are revisited, refined, or abandoned over days and weeks. Because the survey questions on depth- vs. breadth-first workflows focused on divergent exploration, they did not assess whether GenUI also helps users develop ideas more fully or in greater depth. Relatedly, we examined exploration processes and trade-offs rather than evaluating the quality of the final design outcomes produced. Longitudinal studies that track both process and outcome quality would complement our findings.

Finally, our comparison surfaced trade-offs between two coherent GenUI paradigms rather than isolating each design factor independently. Bundling input structure, exploration workflow, and output fidelity made the tools meaningful to use as end-to-end systems, but also made it difficult to attribute observed effects to any single dimension. The tools also differed in maturity: one was a commercial beta product, while the other was our research prototype. In particular, our SVG-based implementation of low-fidelity generation may have influenced factors beyond fidelity itself, such as perceived system capability or instruction-following quality. Future studies could compare tools with more similar levels of polish, or use controlled prototypes that vary one design dimension at a time. Moreover, because our study focused on individual exploration, it does not capture the collaborative nature of UX work; future work should examine how GenUI supports group ideation, coordination, and co-creation.

GenUI promises to democratize UX design exploration for all, yet its very own interface—the UI of GenUI—has remained derivative of existing chat-based AI: starting with an unstructured prompt, delving into a design right away, and generating high-fidelity mockups. In this work, we rethink GenUI designs by invoking a contrastive approach, comparing unstructured vs. structured input, depth- vs. breadth-first workflow, and high- vs. low-fidelity mockup

generation. Juxtaposing such a dichotomy of GenUI design led to a holistic view of the benefits and trade-offs along each of the three aspects, uncovering many a unexpected insight, *e.g.*, the resistance of starting exploration with structured input, the fundamental challenge of previewing UX ideas during exploration, and the reasons behind a overwhelming preference of high over low fidelity. In part, these findings illustrate the inertia for GenUI to break free from its existing UI conventions, and further suggest more calibrated designs to strike a subtle balance between the two sets of dichotomic choices. We see the outcome of this work not only as a path forward for GenUI, but as a broader provocation for how AI-powered creativity tools can be reimaged at their very foundations.

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A Step-by-Step Implementation Details of MINDSPAN

A.1 Generating Design Space

Step 1: Brainstorm multiple ideas

Model: gemini-2.5-flash-lite

Prompt:

SYSTEM:

You are an expert UX strategist and creative thinker.
Generate fundamentally different UI design ideas for the given context.

ASSISTANT:

Brainstorm at least 15 distinct product metaphors or paradigms
to inform the design of this application. Each idea should represent
a different conceptual approach to solving the user's problem.

Inputs:

- Context: {context}
- User: {user}
- Goal: {goal}
- Tasks: {tasks}
- Examples: {examples}

Requirements:

- At least 15 ideas, relevant to the context, user, goal, and tasks
- Draw inspiration from provided examples and user comments,
but do not be constrained by them
- Focus on high-level conceptual approaches, not implementation details

Output Format:

Return a JSON array where each idea object includes:

- idea_id: Unique integer identifier

- idea_name: Short, descriptive name
- description: Brief explanation of how this idea would work
- inspiration: Relation to the context, user, goal, or examples

Constraints:

- Response must be a valid JSON array
- Do not include any text, code fences, or annotations before/after the JSON
- Each idea must be conceptually distinct

Step 2: Identify design dimensions

Model: gemini-2.5-flash-lite

Prompt:

SYSTEM:

You are an expert UX strategist. Create a high-level conceptual design space from the provided divergent ideas, using descriptive, plain language understandable to non-designers (e.g., product managers, software engineers, UX researchers).

ASSISTANT:

Follow these steps:

STEP 1 - Analyze Ideas

- Review all ideas to identify themes, tensions, and patterns
- Highlight fundamental UX-related differences

STEP 2 - Identify Dimensions

- Derive 3 orthogonal design dimensions, each reflecting a key tension, role, or workflow shift
- Each dimension should express a meaningful choice that shapes the user experience
- Write one sentence explaining why each dimension matters

STEP 3 - Create Options

- For each dimension, define 3-5 clearly distinct options
- Each option should represent a different approach along the spectrum
- Provide one-sentence plain-language descriptions

Inputs:

- Divergent Ideas: {divergentIdeas}
- Context: {context}
- User: {user}
- Goal: {goal}
- Tasks: {tasks}
- Examples: {examples}

Requirements:

- Exactly 3 design dimensions, each distinct and relevant to the context
- Each dimension's description should be concise, ideally a question that prompts reflection
- Options must use plain, accessible language and represent meaningful alternatives
- Exclude low-level design details (e.g., styling, information density)
- Use divergent ideas, examples, and user comments as inspiration

Constraints:

- Response must be a valid JSON array
- Do not include text, code fences, or annotations before/after the JSON
- Each dimension and its options must be clearly distinct and relevant

A.2 Generating Specific Designs

Step 1: Generate design concepts

Model: gemini-2.5-flash-lite

Prompt:

Analyze the provided information and generate 5 distinct high-level design concepts that help the user achieve their goal by performing the tasks. Each concept must directly address the following:

Design Parameters: {designParameters}

Output Format:

Return a JSON array where each design object includes:

- design_id: Unique integer identifier
- design_name: A highly descriptive short sentence that conveys the design without needing further explanation
- core_concept: Main design approach, showing how it addresses each parameter (use bullet points)
- detailed_description: A detailed explanation of the concept for a UX designer to implement

Constraints:

- Response must be a valid JSON array
- Do not include text, code fences, or annotations before/after the JSON
- Incorporate user comments to align with preferences and feedback

Step 2: Generate task-wise screen descriptions

Model: gemini-2.5-flash-lite

Prompt:

Analyze the overall design of the application. For each task, generate a 2-3 sentence description of the screens needed to complete it. Each description must specify the screen’s purpose, UI elements, and functionality.

Inputs:

- Overall Design: {overallDesign}
- Tasks: {tasks}

Each screen description should include:

- Purpose: What the screen is for and what users accomplish
- Elements: Key UI components (buttons, forms, lists, navigation, etc.; can be nested objects or arrays)
- Functionality: Actions users can perform and how the screen behaves
- Layout: Organization and positioning of elements
- Interactions: How users navigate to/from this screen and interact

Descriptions must be detailed enough for a UX designer to create the screen.

Step 3: Generate unified screen descriptions

Model: gemini-2.5-flash-lite

Prompt:

You are a UX design assistant helping define the core structure of an application. I will provide several user tasks, each with a sequence of screens. Analyze these task flows and propose a unified set of conceptual screens that support all tasks, while minimizing redundancy and preserving key functionality. Use as few screens as possible.

Output high-level screen concepts, not implementation details. Focus on each screen’s role in the workflow, core elements, and essential interactions. Avoid detailed layouts or exhaustive UI specifications.

For each screen, include:

- title: Short, descriptive name

- purpose: What users achieve and why it is necessary
- core_elements: Major components essential to the screen's purpose (e.g., search bar, content list, form section)
- key_interactions: Main actions users can perform and navigation to/from this screen
- data_notes: (Optional) Data shown or collected, only if essential

Task-Specific Screen Descriptions:
{screenDescriptions}

Requirements:

- Return a JSON array of screen objects that unify overlapping functionality, clarify intent, and remain general enough for low-fi prototyping
- Each screen object must include: title, purpose, core_elements, key_interactions, and optional data_notes
- Response must be a valid JSON array with no text, code fences, or annotations before/after

A.3 Generating UI Code and Taskflows

Step 1: Map screens to each task

Model: gemini-2.5-flash-lite

Prompt:

Given the following tasks and screen descriptions, determine which sequence of screens is needed to complete each task. Use 0-based indices to reference screens.

Tasks:
{tasks}

Screen Descriptions:
{screenDescriptions}

For each task:

- Identify the required screens in order
- Include indices of all necessary screens; each task must use at least one
- Indices cannot exceed the number of available screens
- Typically, a screen does not repeat within the same task
- Each screen must be used by at least one task

Output Format:

Return a JSON object with a "tasksWithScreens" array. Each entry includes:

- task: the task description
- screens: ordered array of screen objects
 - screen_id: index of the screen
 - interaction: brief description of what the user does on this screen to progress to the next

Constraints:

- Response must be a valid JSON object
- Do not include any text, code fences, or annotations before/after the JSON

Step 2: Generate UI code for each screen

Model: gemini-2.5-flash-lite or gemini-2.5-pro

Prompt:

ROLE & GOAL:

You are an expert UI wireframing assistant. Produce one complete, valid SVG that looks like a hand-drawn wireframe of the screen described below.

SCREEN DESCRIPTION:
{screenDescription}

REQUIREMENTS:

- Include every UI element listed in "elements"
- Implement all behaviors in "functionality"
- Reflect user comments and preferences in layout and style

STYLE & FORMAT RULES:

1. Root <svg> must include a proper viewBox
2. All attribute values must be fully quoted (d, x, y, width, height, etc.)
3. No unclosed quotes or stray characters
4. Each <path> must contain a complete d="..."
5. Use stroke and fill attributes; avoid CSS classes
6. Use Comic Sans MS font and grayscale colors only (#000-#FFF) plus transparent fills
7. Output only the SVG markup - no prose, no code fences
8. Apply design principles (accessibility, usability, visual hierarchy) when using user comments

Step 3: Generate task flow

Model: gemini-2.5-flash

Prompt:

Given a task, the SVG UI code for a sequence of screens, and the interaction descriptions for each screen, identify the interactive element(s) the user must engage with to proceed to the next screen.

Task:
{task}

SVG UI Codes:
{uiCodes}

Screen Interactions:
{screenInteractions}

- Requirements:
- For each screen, use the interaction description to determine which element(s) enable transition to the next screen(s)
 - Extract the complete SVG snippet for the identified element(s), including all attributes and nested elements if any
 - Output an array of SVG snippets for each screen in order
 - Be specific: avoid selecting too many elements or overly broad containers
 - Match the interaction description to the correct UI elements

- Constraints:
- Never select the entire <svg>
 - Response must be a valid JSON array with no text before or after
 - Do not mark the response as JSON or use code fences
 - The output array must have the same length as the input uiCodes array
 - If no clear interactive element is found, use an empty array for that screen
 - Focus on elements that align with the described user interaction (e.g., if interaction says "click button", select the button element)

B Logging Details of MINDSPAN

Below is the complete list of JavaScript events that triggered logging in MINDSPAN. We focus on a core set of events that directly capture users' *intentional interactions* with the GenUI tool (e.g., mouse, keyboard, form, and navigation events).

- mousedown: Triggered when a mouse button is pressed down.
- mouseup: Triggered when a mouse button is released.
- click: Fired when a mouse button is pressed and released on an element.
- dblclick: Fired when an element is clicked twice in quick succession.

- `mousemove`: Fired when the mouse pointer moves over an element.
- `contextmenu`: Fired when the user right-clicks to open a context menu.
- `wheel`: Fired when the mouse wheel is rotated.
- `dragstart`: Fired when a drag operation starts.
- `drag`: Fired repeatedly while an element is being dragged.
- `dragend`: Fired when a drag operation ends.
- `drop`: Fired when an element is dropped on a valid target.
- `keydown`: Fired when a key is pressed down.
- `keyup`: Fired when a key is released.
- `keypress`: Fired when a key producing a character value is pressed.
- `input`: Fired when the value of an input or textarea changes.
- `focus`: Fired when an element gains focus.
- `blur`: Fired when an element loses focus.
- `focusin`: Similar to `focus`, but bubbles up.
- `focusout`: Similar to `blur`, but bubbles up.
- `change`: Fired when the value of an input, select, or textarea changes.
- `submit`: Fired when a form is submitted.
- `reset`: Fired when a form is reset.
- `select`: Fired when text inside an input or textarea is selected.
- `copy`: Fired when content is copied to the clipboard.
- `cut`: Fired when content is cut to the clipboard.
- `paste`: Fired when content is pasted from the clipboard.
- `scroll`: Fired when an element's scrollbar is being scrolled.
- `resize`: Fired when the window or an element is resized.
- `visibilitychange`: Fired when the document's visibility changes (e.g., tab hidden).
- `beforeunload`: Fired before a page is unloaded, allowing save prompts.
- `hashchange`: Fired when the URL hash changes.